Report to Huon Valley Council on Waterwatch and Waterbug Blitz activities from July 2022 to June 2023

Introduction.

In mid-2022 the Upper Huon Wildlife Monitoring Landcare Group (UHWMC) received funding from a Huon Valley Council, Natural Resources Management Grant to set up and co-ordinate the collection of water quality data using citizen science based on the popular "Waterwatch" program. The project provides training and support to locally based groups who collect a set of standard water quality parameters (pH, electrical conductivity, temperature and turbidity) from sites on freshwater streams or rivers. The funding also supported training to identify macro-invertebrates as part of the citizen science program the Waterbug Blitz and a public information session.

As an introduction to the issues of water quality, Waterwatch and the Waterbug Blitz the group contributed to two River Health field days (sponsored by the Huon Valley Council) run by Landcare Tasmania in June 2022 at Garden Island Creek and Mountain River. These were followed by a public presentation on "Freshwater Health in the Huon" on Saturday the 30th of July 2022.

The water quality data in this report was collected from the Huon Valley region during the 2nd half of 2022 and 1st half of 2023.

The primary aim of the program is to collect data from fixed sites on a regular basis (monthly to seasonally) to provide information on the variability and trends in water quality data that can be compared to recognised measures of condition used on a statewide or national basis. Additionally, the current data can also be used to compare to historical data collected by state agencies and a similar Waterwatch program run from the late 1990's to the early 2000s.

A second complimentary program to the regular Waterwatch sites is the collection of data over a few days from a much larger number of sites to give a broader understanding of water quality issues in the Huon River catchment. This project is referred to as the "Catchment Crawl" and occurred over four days in late February 2023.

A third component of the project was to provide training to interested residents on the assessment of stream condition using macro-invertebrate composition data collected using the National Waterbug Blitz survey method run by John Gooderham.

Finally, a project is also currently running as a part of a national program using freshwater samples to detect herbicides and pesticides by Deakin University. This project has monthly sampling from April until September from four sites in the Huon, the standard Waterwatch water quality data are also collected in conjunction with the pesticide sampling.

This report is divided into eight parts:

A. Equipment & parameters measured.
B. Information session
C. Waterwatch and Waterbug Blitz
D. Catchment Crawl Feb 23
E. Pesticide sampling.
F. Historic data (Cygnet and Mountain River).
G. Database and sampling sites
H. Results
I. Summary

Appendix A lists all the sites, including site codes and descriptions, latitude and longitude, sampling times, flow and the water quality data (pH, electrical conductivity, temperature and turbidity).

Appendix B lists the participation of group members in activities throughout the year including estimates of volunteer effort in terms of hours.

A. Equipment & parameters measured.

Huon Valley Council ran a significant Waterwatch program at the turn of the century. A lot of the equipment used at that time had been archived and was reassessed for its potential use. Five types of archived equipment where able to be restored or used.

- Hanna Multiprobes (pH/electrical conductivity/temperature)
  These probes are relatively common in Waterwatch programs and are still able to be bought today. Replacement pH probes are also available. Only four of these probes could be restored to use, however the pH readings are subject to variability and require regular calibration or checking against pH paper.
- TDscan probes, (electrical conductivity/temperature)
  Two types of electrical conductivity probes (TDScan 20 & TDScan 3) were restored to use. With six TDScan 20's and four TDScan 3's working in 2022 but two failing in 2023. These probes give very reliable results and only require infrequent calibration.
- Turbidity tubes
  These are a standard method of measurement for Waterwatch and although Huon waters are generally below the detection limit of ~7NTU they provide reliable information of when erosion or resuspension of sediments may be occurring. Six tubes were undamaged.
- Temperature
  Enviro-safe thermometers, four of these where available.
- pH
  Some pHydrion Lo Ion indicator papers (range 5-9) where found to still be usable.

In addition to this pH & electrical conductivity standards and wide range pH papers (pH 1-14) have been purchased by Huon Council or the group to calibrate and provide checks for the probes.

As part of the Catchment Crawl a total of 20 sites were collected for analysis, for soluble and total nutrients by the State Analytical Laboratory (AST). At the same time a sub-set of samples from the twenty sites were analysed using Macherey-Nagel field-based kits for soluble ammonium (visocolor HE test 920006) and phosphate (visocolor HE test 920080). Unfortunately, comparisons between the AST results and the Macherey-Nagel tests found that they were very unreliable at medium to low concentrations of nutrients, which are common in the Huon.

B. Information session "Freshwater Health in the Huon" July 2022

A public presentation on "Freshwater Health in the Huon" was run on Saturday the 30th of July 2022. This presentation had speakers from the group (Simon Roberts) and two other freshwater scientists; Chris Bobbie (retired NRE water quality unit aquatic ecologist) and John Gooderham (Waterbug Blitz trainer and aquatic ecologist) outlining past and present information on water quality from the Huon region, followed by a question-and-answer session.

This information session was supported by Landcare Tasmania. The session was advertised by Landcare Tasmania, Huon Council NRM, The Cygnet, Huon & Channel Classifieds and the Huon News. The Huon news also ran a column in support of the information session. Twenty people attended the session and one watched online. There was a lot of interest generated in the potential for Waterwatch and Waterbug Blitz sessions in the future.

C. Waterwatch and Waterbug Blitz

The first training session in use of equipment, sampling and data recording was at the Huon River at Mosquito Point, Ranelagh on the 22nd of July 2022. From this session two Waterwatch Groups were formed; Russell River (1-2 sites/monthly) and Bakers Creek (2 sites quarterly).

A field based Waterwatch training day was run in the Cygnet area on the 28th Aug 2022 but did not lead to a Waterwatch group.

Two Waterbug Blitz sessions were run in November 2022. The first on the 11th of November at Pelverata and then in Geeveston on the 27th of November. These sessions also introduced Waterwatch water quality testing. In addition, the Russell River group ran an additional two Waterbug Blitz sessions at RUS090 on 11/12/22 and 16/4/23.

Two Waterwatch groups have been formed from interest generated by the Waterbug Blitz sessions (Castle Forbes Bay and Prices Creek with 3 sites each sampled monthly). Training for these groups was provided on site, Castle Forbes Bay in early December 2022 and Prices Creek in February 2023. Both these groups have a field kit with their own equipment and report their results by email. A third group was established in the Port Huon/Geeveston area but is currently only doing sporadic sampling.

In total there are now four active Waterwatch groups doing regular monitoring of nine sites with a further two more sites receiving some sporadic sampling. As a result of the various training sessions a number of other sites where also assessed at least once for both water quality and macroinvertebrates.

D. Catchment Crawl Feb 23

This project has the objective of giving a “snapshot” of the pattern of water quality across a broader spatial area of the Huon River catchment and primarily based on a similar program run in 1997 by the State Government (Bobbi 1998). This project is referred to as the “Catchment Crawl” and was run from the 24th to 27th of February 2023. A total of 38 sites were sampled for the standard Waterwatch parameters with 20 of these sites also sampled for soluble and total nutrients for analysis at Analytical Services Tasmania. Each sampling day concentrated on a region of the Huon defined as the north, east, west and south. On each day two teams of between two and three people sampled four or five sites during the morning or early afternoon, eleven people participated in this program.

The standard water quality data from this project has been incorporated into the general Huon Waterwatch database. The nutrient data have been collated and compared to the values collected in 1996.

E. Pesticide sampling

An Australia wide citizen science program is being run by Deakin University to determine the potential contamination of freshwaters from herbicides and weedicides. As part of these programs the university requested that Waterwatch groups collect samples for analysis from when collected the standard water quality data. The university provided appropriate sampling equipment and instructions as well as all the packaging and pre-paid postage envelopes for the samples to be returned. Sampling occurs once a month from April to September 2023. Four sites are being sampled, one each in the lower reaches of the Huon River, Mountain River, Kermandie River and Agnes Rivulet. The UHWMC and Russell River group collect from two sites each for this project.

The standard water quality data from the first three months of the project has been incorporated into the general Huon Waterwatch database.

F. Cygnet and Mountain River historic sampling

A Waterwatch group was active in the Cygnet area from 1998 to 2010 and generally collected data on a 3-month interval. This data set has been retained as a set of field sheets recording the data collected. In total 24 sites were sampled at different times leading to the collection of 653 sets of site data. At this stage the data have been entered into an excel spreadsheet retaining as much of the information as possible.

In addition, some historic field sheets from the Mountain River Waterwatch group have been obtained, this data is still to be entered into an excel sheet for further analysis.

G. Database and sampling sites

There is currently no central data repository in Tasmania to receive water quality data from Waterwatch activities. The state government database (Water Information System Tasmania) does not currently take data sourced from citizen science activities. All the recently collected data in the Huon is being collated by the UHWMG into excel and distributed via electronic or printed reports.

In total 53 sites had at least one sample collected during the year with 123 sets of water quality measurements collected (Appendix A). In addition, five sites (KEL400, PEL300, STH200, KER600 & RUS090) were sampled and assessed for macroinvertebrates which were submitted to the WaterBug Blitz database.

Sampling occurred through a large area of the Huon River Catchment with results organised into four geographic areas (North, South, East and West) for ease of comparison. Figures 1 to 4 show the position, site code and sample number from each area.

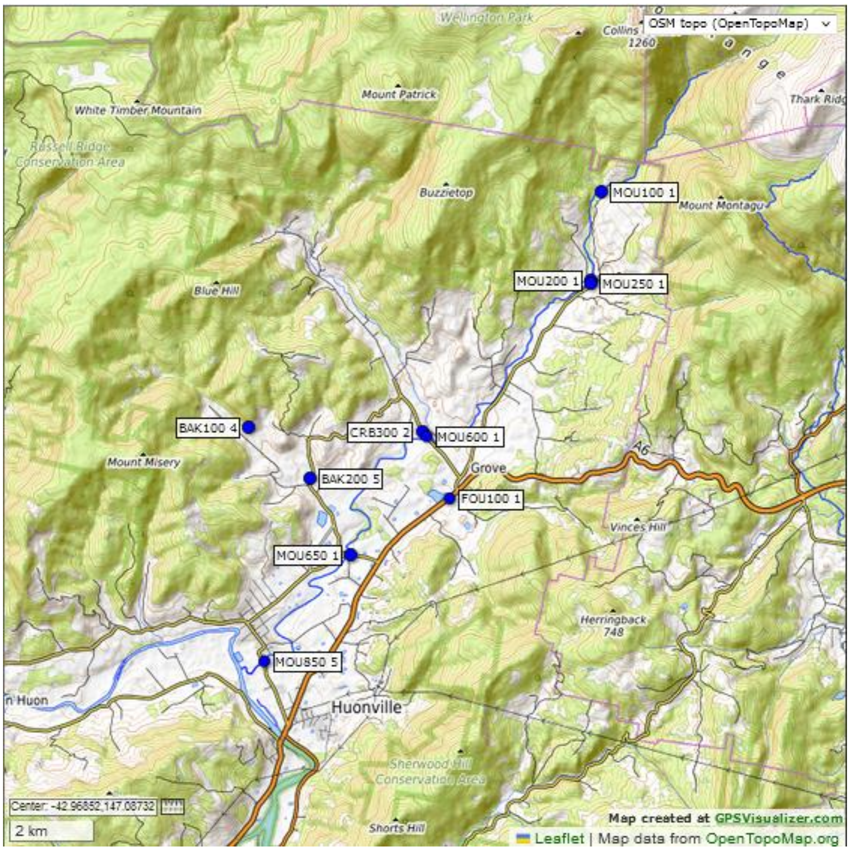


Figure 1. Location and frequency of sampling of Northern area Waterwatch sites monitored between July 2022 and June 2023. Labels show codes for site and number of visits. See Appendix A for details.

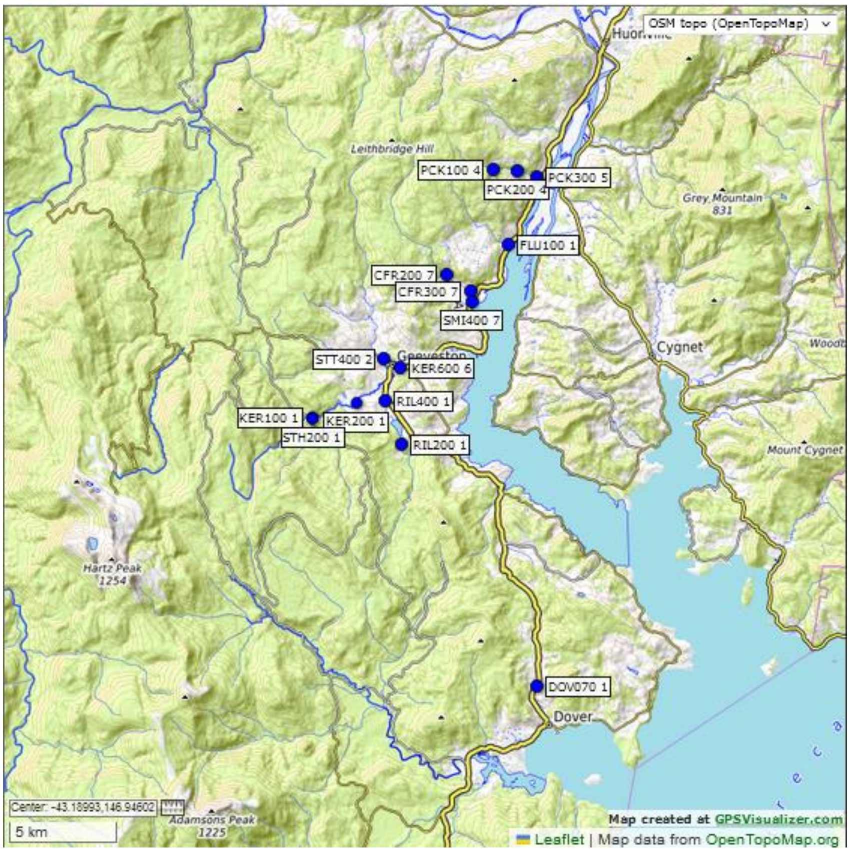


Figure 2. Location and frequency of sampling of Southern area Waterwatch sites monitored between July 2022 and June 2023. Labels show codes for site and number of visits. See Appendix A for details.

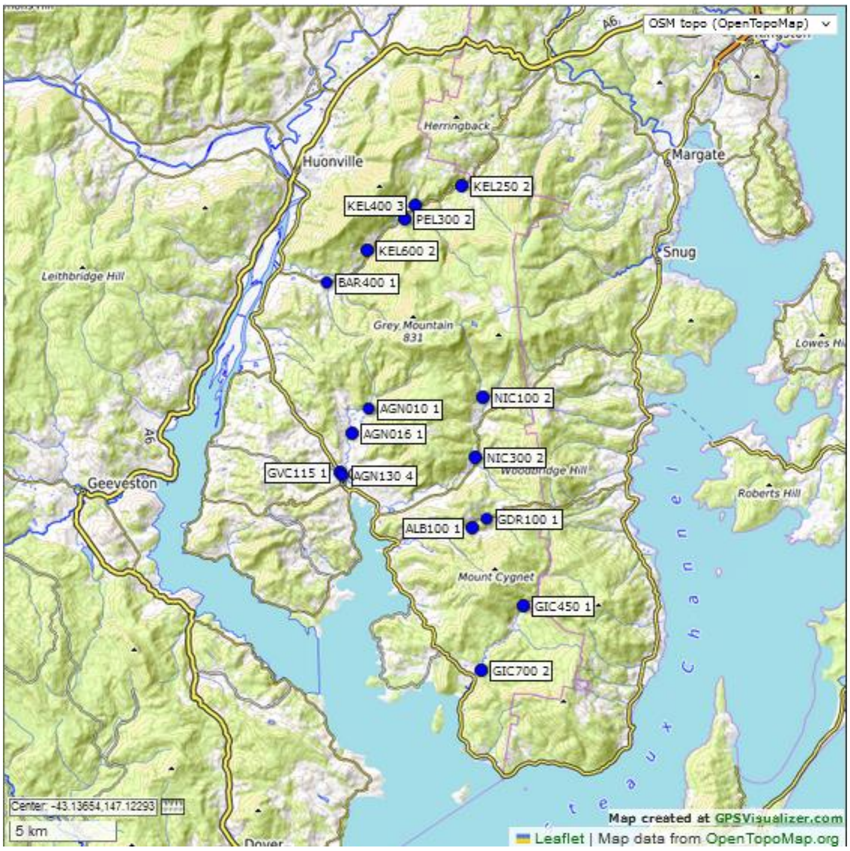


Figure 3. Location and frequency of sampling of Eastern area Waterwatch sites monitored between July 2022 and June 2023. Labels show codes for site and number of visits. See Appendix A for details.

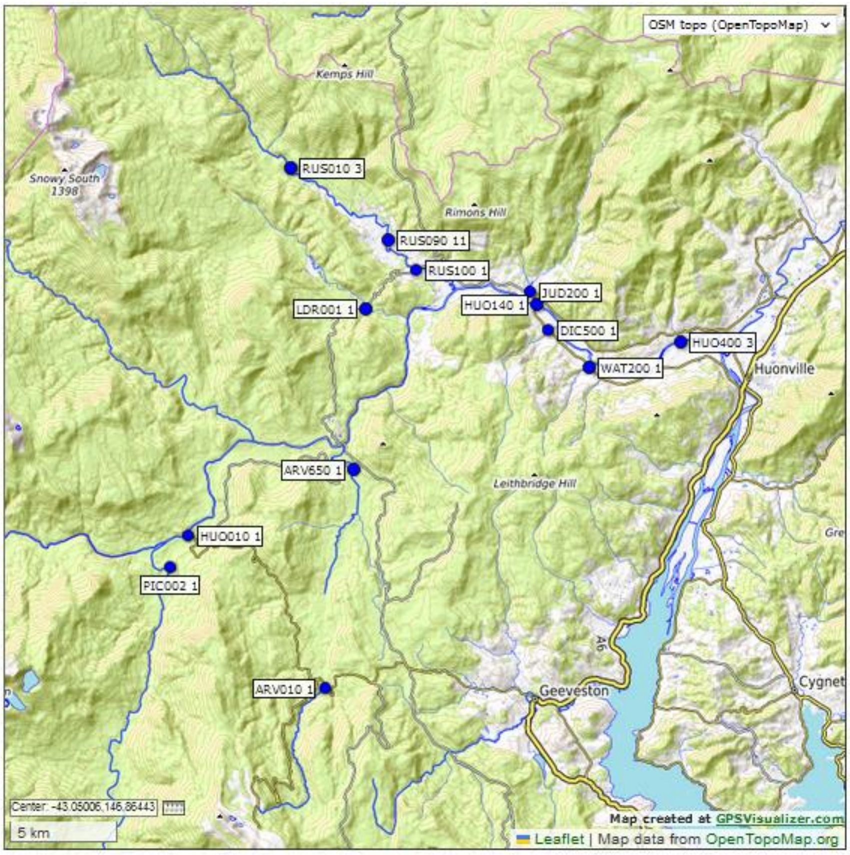
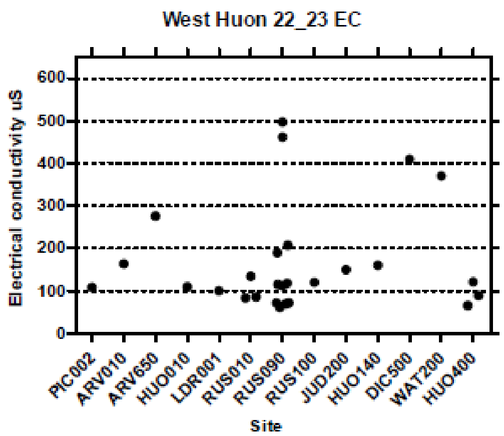


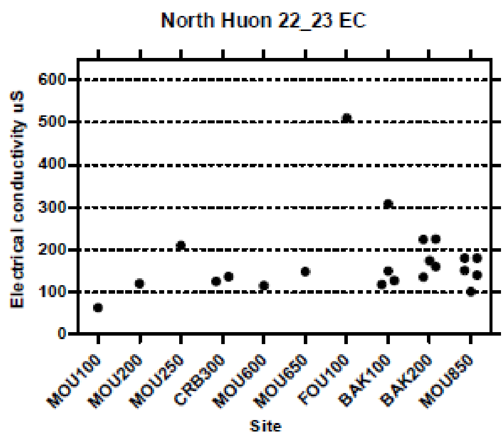
Figure 4. Location and frequency of sampling of Western area Waterwatch sites monitored between July 2022 and June 2023. Labels show codes for site and number of visits. See Appendix A for details.

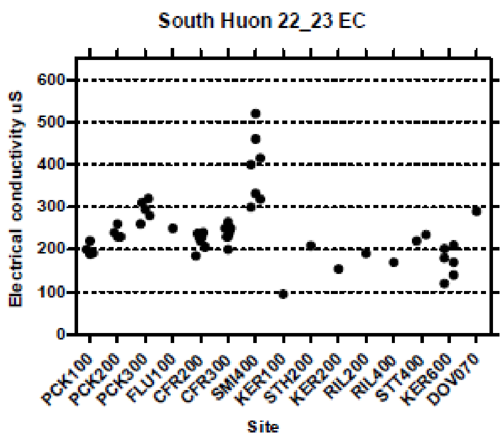
H. Water quality results 2022/23

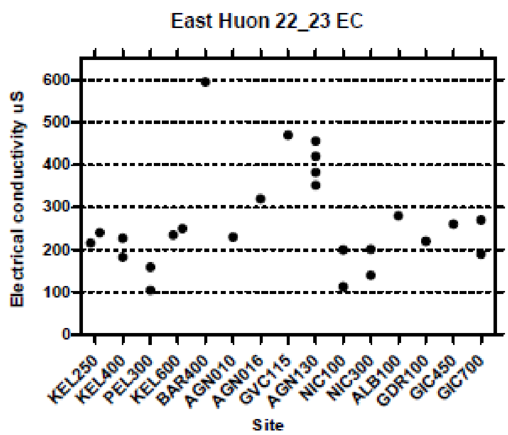
Only a few sites had more than four sampling occasions and at many of these sites sampling started in late in 2022 or early 2023. Currently there is insufficient data to make any definitive statements about trends, annual/seasonal averages or spatial relationships. However, some patterns of parameters are consistent with earlier studies. In particular electrical conductivity and turbidity patterns look broadly in line with previous results that identified more developed catchments having higher conductivity (ie the lower end of Prices Ck, Fourteen Turn Ck, Barretts Ck, Smiths Ck, Dickson and Watsons Ck, Golden Valley Ck and lower Agnes Rivulet) than sites with higher percentages of the catchment in natural or near natural condition. Similarly, turbidity was more likely to be elevated in many of the same waterways and in particular in the waterways of the south area. Note the default value of 4 ntu is used in graphs when the value was below the detection limit of the turbidity tube of 7 ntu.

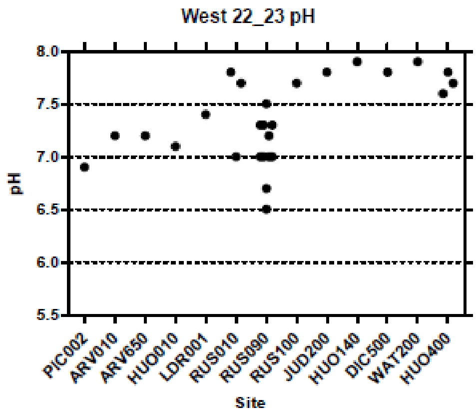
Summary graphs of the distribution of results for pH, electrical conductivity, temperature and turbidity from each site by area (see figures 1 to4) are presented below. Sites are arranged as far as possible geographically moving either downstream (larger site numbers are lower in the system) or from north to south in the particular area of interest (ie Prices Creek is the first in the South area starting from the top site PCK100 and then progressing downstream to PCK300 followed by Fluertys Rivulet and ending with Dover rivulet as the most southern site).

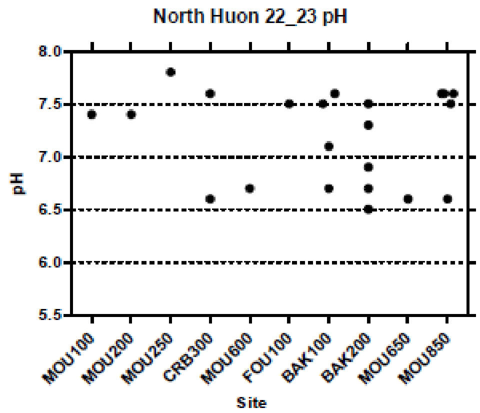


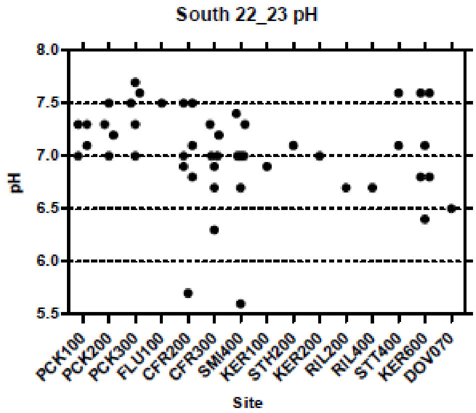


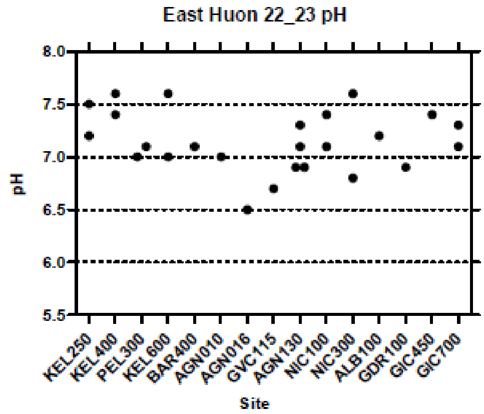


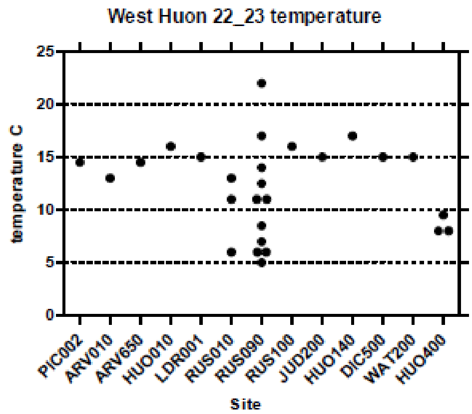


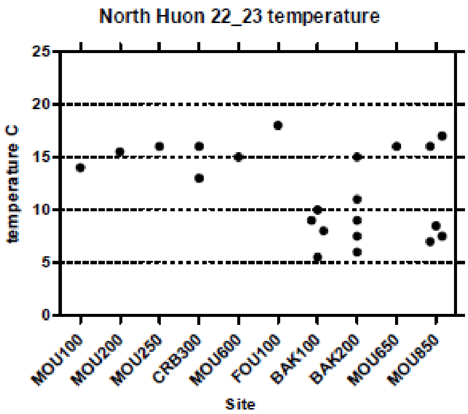

West 22_23 pH
pH
8.0
7.5
7.0
6.5
6.0
5.5
PIC002
ARV010
ARV650
HUO010
LDR001
RUS010
RUS090
RUS100
JUD200
HUO140
DIC500
WAT200
HUO400
Site

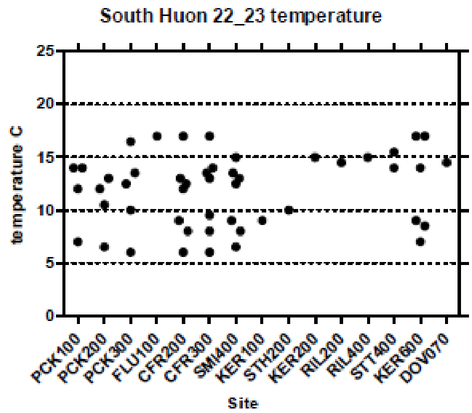

North Huon 22_23 pH
pH
8.0
7.5
7.0
6.5
6.0
5.5
MOU100
MOU200
MOU250
CRB300
MOU600
FOU100
BAK100
BAK200
MOU650
MOU850
Site


South 22_23 pH
pH
8.0
7.5
7.0
6.5
6.0
5.5
PCK100
PCK200
PCK300
FLU100
CFR200
CFR300
SMI400
KER100
STH200
KER200
RIL200
RIL400
STT400
KER600
DOV070
Site


East Huon 22_23 pH
pH
8.0
7.5
7.0
6.5
6.0
5.5
KEL250
KEL400
PEL300
KEL600
BAR400
AGN010
AGN016
GVC115
AGN130
NIC100
NIC300
ALB100
GDR100
GIC450
GIC700
Site


West Huon 22_23 temperature
temperature C
25
20
15
10
5
0
PIC002
ARV010
ARV650
HUO010
LDR001
RUS010
RUS090
RUS100
JUD200
HUO140
DIC500
WAT200
HUO400
Site


North Huon 22_23 temperature
temperature C
25
20
15
10
5
0
MOU100
MOU200
MOU250
CRB300
MOU600
FOU100
BAK100
BAK200
MOU650
MOU850
Site


South Huon 22_23 temperature
temperature C
25
20
15
10
5
0
PCK100
PCK200
PCK300
FLU100
CFR200
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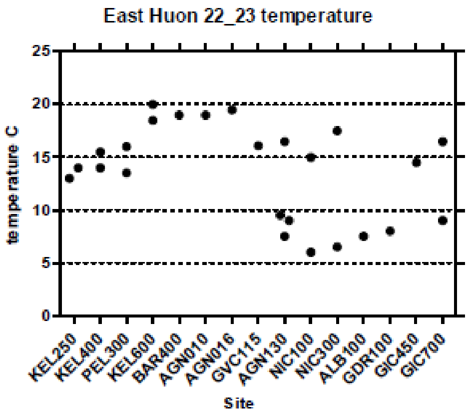
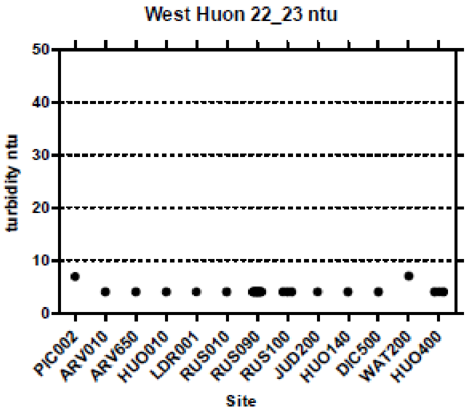
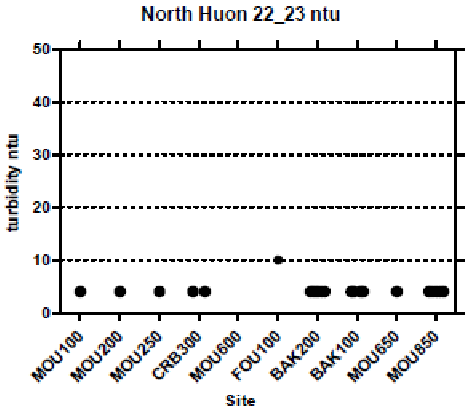
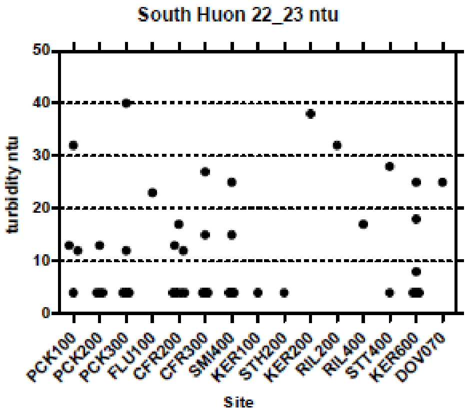
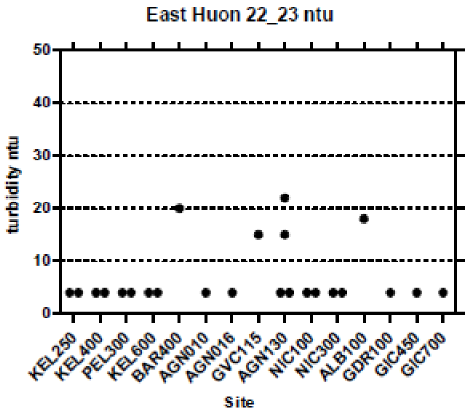
SMI400
KER100
STH200
KER200
RIL200
RIL400
STT400
KER600
DOV070
Site


East Huon 22_23 temperature
temperature C
25
20
15
10
5
0
KEL250
KEL400
PEL300
KEL600
BAR400
AGN010
AGN016
GVC115
AGN130
NIC100
NIC300
ALB100
GDR100
GIC450
GIC700
Site









Nutrient results from 20 sites sampled during the catchment crawl were compared to previous data collected in 1996 and the EPA guidelines for freshwaters in the Huon. The EPA guidelines are not considered a “trigger value” but give guidance as to what the nutrient values are likely to have been prior to significant development or low levels of development (referred to as Slight to Moderately Disturbed- SMD) in the catchments.

For each nutrient (soluble reactive phosphate-SRP; Total phosphate- TP; soluble Ammonium-NH_4; soluble oxidised nitrate- NOx; and total nitrogen- TN) the range of values expected from the EPA SMD default guidelines was compared to the results. If a value was considerably higher than expected in 1 in 20 samples (the 95th percentile) it was considered moderately elevated above expected and marked with bold type in the table. If a value was more than the expected 5th to 95th percentile range of a parameter above the expected default 95th percentile SMD values, it was considered highly elevated and presented in bold type with a grey background in the table.

As stated previously a single value may not be representative of the actual distribution of nutrient concentrations at a site over time, which is clearly shown in the differences in values often seen in the three nutrient values we currently have for each site.

Many values are well within the range expected from the SMD EPA guidelines, particularly SRP, TP, NH_4 and TN. However, there are also many that could be considered highly elevated values indicating that the waterbody may be under nutrient stress. These include four sites with highly elevated SRP and/or highly elevated TP and two sites with highly elevated NH_4. Soluble oxidised nitrogen (NOx) was highly elevated in 13 of the sites and TN was also highly elevated in three of these sites.

Overall based on the single sample from this study and the two from 1997 there is the potential for nutrient stress in a number of waterways in the Huon. Further and broader sampling would help to clarify the extent and frequency of elevated nutrients in a number of these waterways as well as potential sources.

				soluble reactive P (SRP)			Total P (TP)		
code	**date**	**time**	**desc**	**Sum 23**	**Sum 97**	**Wint 97**	**Sum 23**	**Sum 97**	**Wint 97**
AGN130	25/02/2023	10:00	Agnes Rlt, Football Grnd	0.006	0.005	0.011	0.037	na	na
ARV650	27/02/2023	10:11	Arve R, Edwards Rd	0.006	0.002	<0.001	0.016	<0.015	<0.015
BAK200	24/02/2023	9:55	Bakers Ck, Lucaston Rd	**0.140**	0.004	0.005	**0.147**	**0.250**	<0.015
CFR300	26/02/2023	10:45	Castle Forbes Rlt, weir 38 Castle Fo	**0.015**	0.007	0.009	0.056	**0.150**	**0.150**
CRB300	24/02/2023	10:25	Crabtree Rlt, Crabtree Rd	**0.086**	0.005	<0.001	**0.081**	**0.150**	<0.015
DOV070	26/02/2023	10:22	Dover Rlt, Huon Hwy	**0.027**	0.005	Summary	**0.177**	**0.560**	0.350
FOU100	24/02/2023	11:07	Fourteen Turn Ck,Huon Hwy	0.004	<0.001	0.005	0.031	**0.350**	<0.015
GIC450	25/02/2023	10:50	Garden Isl Ck, end of sealed Rd	0.007	0.004	0.004	0.005	<0.015	<0.015
GIC700	25/02/2023	11:16	Garden Isl Ck, Brittains Rd	0.004	0.002	<0.001	0.017	**0.150**	<0.015
GVC115	25/02/2023	10:00	Golden Valley Ck, Slab Rd	0.007	**0.011**	**0.028**	**0.092**	**1.260**	**1.920**
HUO140	27/02/2023	10:32	Huon R, Judbury Bridge	0.002	<0.001	<0.001	0.005	<0.015	<0.015
KEL250	25/02/2024	10:48	Kellaways Ck, Nth Vincents Rd	0.005	0.005	0.005	0.034	**0.150**	<0.015
KER600	26/02/2023	12:15	Kermandie R, Arve Rd	0.004	0.005	0.007	0.066	na	**0.150**
MOU850	24/02/2023	9:50	Mountain R, Wilmot Rd	0.002	0.009	0.004	0.013	**0.250**	<0.015
NIC300	25/02/2023	12:16	Nicholls Rlt, Underwood Rd	0.005	0.005	0.009	0.005	<0.015	**0.200**
PIC002	27/02/2023	11:43	Picton R, Riveaux Rd	0.002	<0.001	<0.001	0.011	<0.015	<0.015
RIL400	26/02/2023	11:23	Rileys Ck, 1st bridge Rileys Ck Rd	0.004	**0.011**	0.009	0.058	**0.510**	**0.300**
RUS100	27/02/2023	11:09	Russell R, Lonnavale Rd	0.002	0.002	<0.001	0.005	<0.015	<0.015
STT400	26/02/2023	12:04	Scotts Rlt, Fourfoot Rd	**0.010**	0.009	**0.014**	**0.087**	**0.250**	**0.350**
WAT200	27/02/2023	9:56	Watsons Ck, Glen Huon Rd	0.008	**0.014**	0.009	0.029	**0.300**	**0.200**

				Ammonium (NH4)			Oxidised N (Nox)			Total N (TN)		
code	**date**	**time**	**desc**	**Sum 23**	**Sum 97**	**Wint 97**	**Sum 23**	**Sum 97**	**Wint 97**	**Sum 23**	**Sum 97**	**Wint 97**
AGN130	25/02/2023	10:00	Agnes Rlt, Football Grnd	0.007	na	0.020	**0.015**	na	na	0.430	na	na
ARV650	27/02/2023	10:11	Arve R, Edwards Rd	0.006	<0.020	<0.020	0.001	<0.010	0.010	0.220	0.260	0.160
BAK200	24/02/2023	9:55	Bakers Ck, Lucaston Rd	0.006	<0.020	<0.020	**0.012**	**0.060**	**0.090**	0.260	0.470	0.370
CFR300	26/02/2023	10:45	Castle Forbes Rlt, weir 38 Castle Fo	0.017	<0.020	0.020	**0.360**	**0.020**	**0.020**	**1.100**	0.310	0.260
CRB300	24/02/2023	10:25	Crabtree Rlt, Crabtree Rd	0.003	<0.020	<0.020	**0.033**	**0.060**	**0.040**	0.160	0.260	0.160
DOV070	26/02/2023	10:22	Dover Rlt, Huon Hwy	**0.170**	**0.060**	**0.070**	**0.500**	**0.060**	**0.240**	**2.200**	0.840	0.940
FOU100	24/02/2023	11:07	Fourteen Turn Ck,Huon Hwy	0.014	**0.080**	<0.020	**0.051**	<0.010	0.010	0.510	0.520	0.210
GIC450	25/02/2023	10:50	Garden Isl Ck, end of sealed Rd	0.005	<0.020	<0.020	**0.110**	**0.040**	**0.170**	0.360	0.260	0.260
GIC700	25/02/2023	11:16	Garden Isl Ck, Brittains Rd	0.006	<0.020	<0.020	0.003	<0.010	<0.010	0.330	0.370	0.160
GVC115	25/02/2023	10:00	Golden Valley Ck, Slab Rd	0.018	**0.160**	**0.050**	**0.290**	**0.300**	**2.000**	**1.300**	**1.520**	**1.940**
HUO140	27/02/2023	10:32	Huon R, Judbury Bridge	0.003	<0.020	<0.020	0.003	<0.010	0.010	0.190	<0.15	0.210
KEL250	25/02/2024	10:48	Kellaways Ck, Nth Vincents Rd	0.007	<0.020	<0.020	**0.097**	**0.040**	**0.040**	0.490	0.370	0.260
KER600	26/02/2023	12:15	Kermandie R, Arve Rd	0.011	na	na	**0.510**	**0.040**	**0.120**	**1.400**	0.420	0.420
MOU850	24/02/2023	9:50	Mountain R, Wilmot Rd	0.003	<0.020	0.020	0.003	**0.020**	0.010	0.230	0.210	0.160
NIC300	25/02/2023	12:16	Nicholls Rlt, Underwood Rd	0.005	0.020	<0.020	**0.089**	**0.120**	**0.160**	0.330	0.420	0.310
PIC002	27/02/2023	11:43	Picton R, Riveaux Rd	0.008	<0.020	<0.020	0.001	<0.010	<0.010	0.260	<0.15	0.210
RIL400	26/02/2023	11:23	Rileys Ck, 1st bridge Rileys Ck Rd	**0.150**	0.030	0.020	**0.140**	**0.090**	**0.160**	**1.100**	0.790	0.790
RUS100	27/02/2023	11:09	Russell R, Lonnavale Rd	0.003	<0.020	<0.020	0.001	<0.010	**0.020**	0.050	0.160	<0.15
STT400	26/02/2023	12:04	Scotts Rlt, Fourfoot Rd	0.007	0.020	0.030	**0.230**	**0.160**	**0.200**	**1.200**	0.580	0.520
WAT200	27/02/2023	9:56	Watsons Ck, Glen Huon Rd	0.003	<0.020	<0.020	**0.050**	**0.020**	**0.200**	0.530	0.310	0.470

I. Summary

There has been a high level of interest and participation in the water quality project run by the UHWMG. Many people have committed their time to come to training sessions and be involved in ongoing data collection. Overall, in 2022/23 we have:

- Run a general information meeting.
- Run four Waterwatch training sessions.
- Run Two Waterbug Blitz training sessions.
- Initiated Two Waterbug blitz monitoring days.
- Established four Waterwatch groups sampling nine sites regularly.
- Run a “Catchment Crawl” monitoring 38 sites.
- Supported a National Pesticide monitoring program at four sites.
- Collated and analysed historic data (ongoing)
- Established a Database and distributed the water quality data collected in the program.

Key outcomes from the results, indicate:

- Both regular monitoring and broadscale single site monitoring provide useful data for making contemporary assessments of the health of waterways and to make comparisons both spatially and to historic data.
- A longer set of data is required to make definitive statements about the water quality of freshwater systems in the Huon; however,
- The current data indicates that water quality in many of the lowland more developed catchments may not have improved since the start of the century; and
- That a number of waterways may be subject to nutrient stress.

Although there are many challenges to water quality monitoring it is inspiring to see the level of commitment and interest in the program. In the coming year we hope to support more Waterwatch groups (equipment allowing), establish a management team, rerun the Catchment Crawl, and provide further training in the Waterwatch and the Waterbug Blitz program for established and new groups. Both the Huon Valley Council Natural Resource Management team and Landcare Tasmania have been instrumental in providing support for this program. Similarly, all participants in the group’s activities and in particular the members of the local Waterwatch groups and Catchment Crawl team have been crucial to the program’s success.

Appendix A. Site codes, latitude and longitude, site descriptions, sampling times and water quality data Huon catchment 2022/23.

name	latitude	longitude	description	date	time	flow	EC	pH	turb tube	temp
ALB100	-43.17802	147.1493	Albert Ck, Coal Mine Rd	28/08/2022	12:30pm	medium	280	7.2	18	7.5
AGN010	-43.12712	147.0907	Agnes Rlt, O'Brians Rd	25/02/2023	10:50am	low	230	7	4	19
AGN016	-43.1391	147.0806	Agnes Rlt, Woodcock Rd	25/02/2023	12:08pm	low	310	6.5	4	19.5
AGN130	-43.15662	147.0751	Agnes Rlt, Football Grnd	25/02/2023	10:00am	low	352	7.1	4	16.5
				27/04/2023	7:30am	low	382	7.3	4	9
				27/05/2023	14:00pm	medium	456	6.9	15	9.5
				19/06/2023	13:20pm	low	420	6.9	22	7.5
ARV010	-43.15872	146.8071	Arve R, Arve Rd picnic area	27/02/2023		low	164	7.2		13
ARV650	-43.06725	146.8233	Arve R, Edwards Rd	27/02/2023	10:10am	low	276	7.2	4	14.5
BAK100	-42.96867	147.0369	Bakers Ck, Adyah farm	29/07/2022	11:20am	medium	118	7.5	4	5.5
				4/11/2022	10:00am	medium	308	7.1	4	9
				3/02/2023	9:45am	medium	127	7.6	4	10
				5/05/2023	9:30am	low	150	6.7	4	8
BAK200	-42.97938	147.0545	Bakers Ck, Lucaston Rd	29/07/2022	10:35am	medium	225	7.3	4	6
				4/11/2022	9:20am	medium	135	6.9	4	9
				3/02/2023	9:00am	medium	160	7.5	4	11
				24/02/2023	9:55am	low	174	6.5	4	15
				5/05/2023	9:00am	low	224	6.7	4	7.5
BAR400	-43.07608	147.0665	Barretts Ck, Pelverata Rd	29/01/2023	13:49pm	litte flow	595	7.1	20	19
CFR200	-43.1261	146.9587	Castle Forbes Rlt, Harwoods Rd	7/01/2023	9:15am	low	206	6.8	4	13
				4/02/2023	9:20am	low	238	6.9	4	12.5
				26/02/2023	10:30am	low	220	7.1	17	17
				7/03/2023	9:15am	low	230	7.5	12	12
				1/04/2023	9:00am	low	240	5.7	4	9
				6/05/2023	9:00am	low	230	7.5	4	6
				3/06/2023	8:45am	low	185	7	13	8
CFR300	-43.13287	146.9723	Castle Forbes Rlt, weir	7/01/2023	8:30am	low	239	6.7	4	14
				4/02/2023	9:00am	low	265	6.9	4	13.5
				26/02/2023	10:45am	low	230	7.2	27	17
				7/03/2023	8:30am	low	230	7.3	4	13
				1/04/2023	8:30am	low	250	6.3	4	9.5
				6/05/2023	8:30am	little flow	250	7	4	6
				3/06/2023	8:20am	little flow	200	7	15	8
CRB300	-42.96955	147.0868	Crabtree Rlt, Crabtree Rd	3/02/2023	10:40am	medium	125	7.6	4	13
				24/02/2023	10:25am	low	136	6.6	4	16
DIC500	-43.00928	146.9344	Dickensons Ck, Glen Huon Rd	27/02/2023	10:15am	low	410	7.8	4	15
DOV070	-43.29767	147.0102	Dover Rlt, Huon Hwy	26/02/2023	10:22am	low	290	6.5	25	14.5
FLU100	-43.11335	146.9939	Fluertys Rlt, 4 Murrells Rd	26/02/2023	11:40am	low	250	7.5	23	17
FOU100	-42.98365	147.0941	Fourteen Turn Ck,Huon Hwy	24/02/2023	11:07am	little flow	510	7.5	10	18
GDR100	-43.17455	147.1578	Gardners Ck, Bluegum Rd	28/08/2022	12:00pm	medium	220	6.9	4	8
GIC450	-43.2109	147.1793	Garden Isl Ck, end of sealed Rd	25/02/2023	10:50am	low	260	7.4	4	14.5
GIC700	-43.2377	147.1547	Garden Isl Ck, Brittains Rd	18/06/2022	14:20pm	medium	190	7.1	4	9
				25/02/2023	11:15am	low	270	7.3		16.5
GVC115	-43.15527	147.0741	Golden Valley Ck, Slab Rd	25/02/2023	10:00am	low	470	6.7	15	16.1
HUO010	-43.09504	146.7289	Huon R, Tahune bridge	27/02/2023	12:17pm	low	109	7.1	4	16
HUO140	-42.99877	146.9282	Huon R, Judbury Bridge	27/02/2023	10:32am	low	160	7.9	4	17
HUO400	-43.0143	147.0108	Huon R, Smiths Rd	26/04/2023	10.00 am	low	89	7.8	4	9.5
				19/05/2023	10.30 am	medium	65	7.6	4	8
				15/06/2023	10.05 am	medium	121	7.7	4	8
JUD200	-42.9932	146.9243	Judds Ck, Lonnavale Rd	27/02/2023	10:45am	low	150	7.8	4	15

Waterwatch & Waterbug Blitz report Huon Catchment 2022/23

name	latitude	longitude	description	date	time	flow	EC	pH	turb tube	temp
KEL250	-43.0354	147.1433	Kellaways Ck, Nth Vincents Rd	28/01/2023	10:30am	medium	216	7.5	4	13
				25/02/2023	10:48am	low	240	7.2	4	14
KEL400	-43.044	147.1175	Kellaways Ck, Halls Track Rd	11/11/2023	10:30am	medium	183	7.4	4	14
				28/01/2023	11:20am	medium	227	7.6	4	15.5
KEL600	-43.0624	147.0896	Kellaways Ck, below Lords Rd	29/01/2023	14:34pm	medium	235	7	4	20
				26/02/2023	11:09am	low	250	7.6	4	18.5
KER100	-43.1858	146.8817	Kermandie R, u/s South Ck	27/11/2022	8:30am	medium	95	6.9	4	9
KER200	-43.1797	146.9072	Kermandie R, McKibbens Rd	26/02/2023	11:45am	medium	154	7	38	15
KER600	-43.165	146.9318	Kermandie R, Arve Rd	27/11/2022	10:30am	medium	210	7.1	4	14
				4/02/2023	9:50am	medium	202	7.6	4	17
				25/02/2023	12:15pm	high	180	7.6	25	17
				26/04/2023	9:20am	low	170	6.8	4	9
				29/05/2023	13:30pm	medium	120	6.8	18	8.5
				28/06/2023	13:00pm	medium	140	6.4	8	7
LDR001	-43	146.8303	Little Denison R, Denison Rd	27/02/2023	11:30am	low	100	7.4	4	15
MOU100	-42.9193	147.1378	Mountain R, end of Mtn R Rd	15/11/2023	14:30pm	medium	63	7.4	4	14
MOU200	-42.9378	147.1346	Mountain R, Bennetts Rd	24/02/2023	10:30am	low	120	7.4	4	15.5
MOU250	-42.9386	147.1346	Grundys Ck, Mtn R Rd	24/02/2023	10:40am	low	210	7.8	4	16
MOU600	-42.9706	147.0877	Mountain R, Crabtree Rd	24/02/2023	10:46am	low	115	6.7		15
MOU650	-42.9954	147.0662	Mountain R, Lollara Rd	24/02/2023	11:30am	low	148	6.6	4	16
MOU850	-43.0177	147.0413	Mountain R, Wilmot Rd	4/02/2023	11:15am	medium	151	7.6	4	16
				24/02/2023	9:50am	low	180	7.6	4	17
				26/04/2023	8:20am	low	140	6.6	4	8.5
				19/05/2023	9:00am	low	180	7.6	4	7
				15/06/2023	10:50am	low	101	7.5	4	7.5
NIC100	-43.1243	147.1558	Nicholls Rlt, Smiths Rd	28/08/2022	10:20am	medium	113	7.1	4	6
				25/02/2023	11:50am	low	200	7.4	4	15
NIC300	-43.149	147.1512	Nicholls Rlt, Underwood Rd	28/08/2022	11:20am	medium	140	6.8	4	6.5
				25/02/2023	12:15am	low	201	7.6	4	17.5
PCK100	-43.0822	146.9854	Prices Ck, Rankins Rd	15/02/2023	12:30pm	little flow	192	7.3	4	14
				11/03/2023	12:00pm	little flow	190	7.3	13	14
				12/04/2023	11:20am	low	220	7.1	12	12
				18/05/2023	13:20pm	little flow	200	7	32	7
PCK200	-43.0831	146.9993	Prices Ck, 124 New Rd	15/02/2023	11:30am	little flow	230	7.5	4	13
				11/03/2023	11:40am	little flow	230	7.2	4	12
				12/04/2023	11:05am	low	260	7.3	4	10.5
				18/05/2023	13:10am	little flow	240	7	13	6.5
PCK300	-43.0853	147.0103	Prices Ck, 26 New Rd	15/02/2023	11:00am	little flow	295	7.6	4	12.5
				26/02/2023	11:10am	medium	260	7.7	40	16.5
				11/03/2023	11:20am	little flow	280	7.5	4	13.5
				12/04/2023	10:50am	low	310	7.3	4	10
				18/05/2023	12:50pm	little flow	320	7	12	6
PEL300	-43.0496	147.1111	Pelverata Ck, Pelverata Rd	11/11/2022	9:40am	medium	105	7	4	13.5
				28/01/2023	11:40am		159	7.1	4	16
PIC002	-43.108	146.7181	Picton R, Riveaux Rd	27/02/2023	11:45am	low	108	6.9		14.5
RIL200	-43.1969	146.9326	Rileys Ck, above dam	26/02/2023	12:00pm	low	191	6.7	32	14.5
RIL400	-43.1789	146.9238	Rileys Ck, 1st bridge Rileys Ck Rd	26/02/2023	11:30am	low	170	6.7	17	15

Waterwatch & Waterbug Blitz report Huon Catchment 2022/23

name	latitude	longitude	description	date	time	flow	EC	pH	turb tube	temp
RUS010	-42.94137	146.7878	Russell R, Russell Road	11/12/2022	9:42am	medium	134	7	4	11
				20/05/2023	14:50pm	medium	83	7.8	4	6
				16/06/2023	14:30pm	medium	86	7.7	4	13
RUS090	-42.9713	146.8436	Russell R, Orana 898 Lonnavale R	31/07/2022	15:00pm	low	118	7.2	4	5
				27/08/2022	2.30 pm	medium	115	7	4	6
				3/10/2022	12.40 pm	low	497	7	4	8.5
				4/11/2022	3.45pm	low	207	7	4	11
				11/12/2022	4.10pm	medium	112	7	4	12.5
				13/01/2023	2.15pm	low	462	7.3	4	22
				12/02/2023	4pm	low	190	6.7	4	17
				18/03/2023	1.05pm	low	69	6.5	4	14
				16/04/2023	3.00pm	medium	72	7.5	4	11
				20/05/2023	3.32pm	low	72	7.3	4	6
				16/06/2023	4.00 pm	low	61	7.3	4	7
RUS100	-42.98383	146.8593	Russell R, Lonnavale Rd	27/02/2023	11:10am	low	120	7.7	4	16
SMI400	-43.1375	146.9729	Smiths Ck, Huon Hwy	10/12/2022	3:30pm	low	318	7.3	4	15
				7/01/2023	8:00am	little flow	332	6.7	4	12.5
				4/02/2023	8:10am	little flow	415	7	4	13.5
				7/03/2023	8:30am	little flow	300	7.4	25	13
				1/04/2023	8:10am	little flow	460	5.6	4	9
				6/05/2023	8:00am	little flow	520	7	4	6.5
				3/06/2023	8:00am	little flow	400	7	15	8
STH200	-43.1859	146.8819	South Ck, at Kermandie R	27/11/2023	8:45am	medium	209	7.1	4	10
STT400	-43.16127	146.9225	Scotts Rlt, Fourfoot Rd	4/02/2023	10:20am	medium	235	7.6	4	14
				26/02/2023	12:04pm	medium	220	7.1	28	15.5
WAT200	-43.02492	146.9579	Watsons Ck, Glen Huon Rd	27/02/2023	10:00am	medium	370	7.9	7	15

Appendix B. Participation in Waterwatch/Waterbug Blitz activities in the Huon catchment 2022/23.

Event/activity	date/ start	frequency	participants	Duration (hrs)	hrs (total)
Information session Freshwater Health in the Huon	30/7/22	once	22	5hrs	110
Mosquito point- water quality training	22/7/22	once	6	3hrs	18
Bakers Ck-Waterwatch	29/7/22	quarterly x4	2	1	8
Russell River- Waterwatch	31/7/22	monthly x 11	2	0.6	13
Nicholls Rivulet- water quality training	28/8/22	once	3	4	12
Waterbug Blitz- Pelverata	11/11/22	once	6	5	30
Waterbug Blitz- Geeveston	27/11/22	once	15	4	60
water quality training- Castle Forbes Bay	10/12/22	once	3	3	9
Waterbug Blitz- Russell R (RUS090)	11/12/23	once	4	4	16
Castle Forbes Bay- Waterwatch	4/2/23	monthly x 6	2	1.5	18
water quality training- Prices Ck	15/2/23	once	3	3	9
Catchment Crawl	25-28/2/23	once	4-5day	4	18
Prices Ck-Waterwatch	11/3/23	monthly x 3	2	1.5	9
Waterbug Blitz- Russell R (RUS090)	16/4/23	once	5	4	20
Pesticide monitoring- Four sites	26/4/23	Monthly x 3	2	3	18
				Total	368

Participants in training sessions or waterwatch groups-

Ben E
Bridget W
Cameron S
Carol H
Cathy T
Chris B
Christie L
Desley K
Emmaline K
Gav J
Hayden P
Helen E
Ingrid H
Jane E
John G
Laura S
Lorrie H
Michael M
Michelle J
Mike P
Neralee H
Ro A-H
Rick P
Silver M
Simon R
Stu S,
Stephen W
Tony Mckennie
Warren H